

llmXive Automated Review of ABot-Earth 0.5: Generative 3D Earth Model

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: [qwen.qwen3.5-122b](#)

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several strong quantitative and comparative claims that are not fully supported by the provided evidence or contain internal inconsistencies.

First, the primary quantitative claim in Section 5.1 (“Generative Fidelity”) asserts that the method achieves a state-of-the-art FID of 16.1, a “substantial improvement” over the previous best of 69.5. However, the caption of Table 1 (tab:conditions_results) explicitly admits that “FID/KID values for baselines are computed using different GT sets than ours” and that “poses/viewpoints used for evaluation differ.” This directly contradicts the text’s assertion of a direct, fair comparison. The text frames this as a definitive SOTA result, while the table admits the metrics are “for reference only.” This is a significant claim-accuracy mismatch; the authors cannot claim a direct performance lead when the evaluation conditions are non-comparable.

Second, the deployment section (Section 4.1) contains a minor arithmetic ambiguity. It states a single inference pass covers “1.6km $\times$ 1.6km (2.56km²)” and represents a “64-fold increase in area” compared to “200m $\times$ 200m training tiles.” While $2.56 / 0.04 = 64$, the text phrasing “1.6km $\times$ 1.6km” could be misread as a linear dimension comparison if not carefully parsed. More importantly, the claim of “Infinite” coverage in Table 2 (tab:system_comparison_merged) is an overstatement of the current system’s state. The introduction and deployment sections clarify the system currently covers “over 300 cities” and is limited by a 1.6km² tile size. “Infinite” implies no current bounds, which is factually incorrect for the deployed system described.

Finally, the qualitative comparison in Section 5.2.1 regarding Ireland relies on the claim that ABot-Earth generates a “plausible 3D scene” where Google Earth fails. While Figure 3 provides visual evidence, the text does not provide any quantitative or specific qualitative criteria (beyond “plausible”) to support the assertion that the generated scene is accurate to the real world, rather than a hallucination. Given the paper’s focus on “real-world geospatial complexity,” a claim of “plausibility” without ground-truth verification for that specific region is a load-bearing claim that lacks sufficient evidentiary backing in the text.

These issues are primarily matters of overstatement and mismatched evidence rather than fundamental scientific flaws, but they undermine the credibility of the paper’s central performance claims.

Action items.

- **[writing]** The paper makes several strong quantitative and comparative claims that are not fully supported by the provided evidence or contain internal inconsistencies. First, the primary quantitative claim in Section 5.1 (“Generative Fidelity”) asserts that the method achieves a state-of-the-art FID of 16.1, a “substantial improvement” over the previous best of 69.5. However, the caption of Table 1 (tab:conditions_results) explicitly admits that “FID/KID values for baselines are computed using different G

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens

runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

The figure attempts to compare two models across five metrics but fails due to mixed scales (percentages vs. 0-5 scores) and undefined qualitative axes, making the visual comparison scientifically invalid.

Figure 2

The figure is a clear Google Earth screenshot but fails to demonstrate the generative capabilities claimed in the paper title, serving only as a reference image without a comparative generated output or sufficient descriptive context.

Figure 3

The figure effectively visualizes the global scope of the dataset, but the caption contains a grammatical error with a missing subject, and the map lacks a legend to explain the color coding of the data points.

Figure 4

Figure 4 is a clear and well-structured pipeline diagram that effectively visualizes the data processing workflow described in the caption. All stages, inputs, and outputs are legible, and the logical flow from data collection to training data is easy to follow.

Figure 5

Figure 5 is a clear, self-contained schematic that effectively illustrates the three primary data sources (Satellite, Aerial, Urban) and their respective roles in the reconstruction pipeline. The visual hierarchy and labels align perfectly with the caption's description of complementary viewpoint coverage.

Figure 6

The figure displays a static tiling grid on a globe but fails to visually represent the 'production pipeline' or 'parallel inference' described in the caption, creating a disconnect between the visual content and the scientific claim.

Figure 7

The figure effectively illustrates the hierarchical quadtree structure and LOD levels (Z14-Z19) with clear visual progression. The internal legend and labels are sufficient to explain the data sources and subdivision rules without needing external cross-references.

Figure 8

The figure fails to illustrate the ‘production pipeline’ described in its caption, showing only a static tiling grid. Additionally, the image and caption are identical to Figures 6 and 9, indicating a likely duplication error.

Figure 9

The figure fails to support its caption’s claim of illustrating a production pipeline or parallel inference, showing only a static tiling grid. Additionally, the image and caption are identical to Figures 6 and 8, indicating a likely duplication error.

Figure 10

The figure fails to substantiate the caption’s claim of demonstrating Level-of-Detail (LOD) rendering or seamless global-to-street exploration, as it presents only a single static aerial view without comparative scales or transition indicators.

Figure 11

The figure displays landmark integration results, but the caption contains significant inaccuracies regarding the viewing angles (claiming top-down views where oblique views are shown) and potentially misidentifies or misdescribes the content of the third and fourth rows.

Action items.

- **[science]** Figure 1: The ‘Efficiency’ axis uses qualitative labels (‘High’/‘Low’) while others use quantitative scales (0-5), and the data points are not aligned with the grid lines (e.g., the blue point is at the outer edge while the label ‘High’ is near the center), making the metric undefined and the chart unreadable.
- **[science]** Figure 1: The ‘Country/Region Coverage’ axis uses a percentage scale (0-100%) while the other four axes use a 0-5 scale; mixing these incompatible units on a single radar chart without normalization renders the geometric comparison of the polygons meaningless.
- **[science]** Figure 2: The image is a standard Google Earth screenshot (indicated by the watermark) rather than a generative output from the ‘ABot-Earth’ model. As a standalone figure, it fails to demonstrate the paper’s core claim of generating 3D Earth models, serving only as a reference or input without a corresponding generated comparison.
- **[writing]** Figure 2: The caption is insufficient, providing only the source (‘Google Earth’) and location (‘New Zealand’) without describing the specific view parameters (pitch, yaw,

zoom) or the purpose of the image in the context of the study.

- **[writing]** Figure 3: The caption contains a grammatical error and missing subject ('We are unveiling , a generative...'); the comma should be removed and the model name inserted.
- **[science]** Figure 3: The map displays blue and orange dots but lacks a legend or key to define what these colors represent (e.g., training vs. test, or specific data sources).
- **[science]** Figure 6: The caption claims the figure shows a 'tile-based production pipeline' and 'parallel inference,' but the image only displays a static globe with a grid overlay and a red line. It lacks any visual representation of a pipeline, data flow, or inference process.
- **[writing]** Figure 6: The image title 'Global Equal-Area Tiling Scheme' is not reflected in the caption, which describes a 'production pipeline' instead. The caption fails to explain the red line or the specific grid structure shown.
- **[science]** Figure 8: The caption claims to show a 'tile-based production pipeline' and 'parallel inference,' but the image displays a static global tiling grid without any visual representation of the pipeline, data flow, or inference process.
- **[science]** Figure 8: The image is identical to Figure 6 and Figure 9, which share the exact same caption text, suggesting a copy-paste error where the specific content for Figure 8 is missing or duplicated.
- **[science]** Figure 9: The caption claims to show a 'tile-based production pipeline' and 'parallel inference,' but the image displays a static global tiling grid without any visual representation of the pipeline, inference process, or parallelism.
- **[science]** Figure 9: The image is identical to Figure 6 and Figure 8, which share the exact same caption text, suggesting a copy-paste error where the specific content for Figure 9 is missing or duplicated.
- **[science]** Figure 10: The caption claims to show 'LOD rendering' enabling 'seamless exploration from global to street-level views,' but the image displays a single static aerial view of a city without any visual evidence of multi-scale LOD transitions or global context.
- **[writing]** Figure 10: The image lacks any scale bar, compass, or coordinate reference to identify the specific location shown, making the 'global' context claim unverifiable.
- **[science]** Figure 11: The caption claims 'Top-down views are in the leftmost column,' but the images in the first column are clearly oblique aerial views, not top-down (nadir) views. The perspective is slanted, contradicting the description.
- **[writing]** Figure 11: The caption states 'From top to bottom: Eiffel Tower, Colosseum, US Capitol, and Arc de Triomphe,' but the visual content of the rows does not match this order. The first row shows the Eiffel Tower, the second the Colosseum, the third the US Capitol, and the fourth the Arc de Triomphe. However, the third row (US Capitol) appears to be a different building or a significantly distorted version compared to the others, and the fourth row (Arc de Triomphe) is actually the Arc de Triomphe,

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally well-structured for a technical audience, but it relies on several subfield-specific acronyms and terms that are not defined at their first occurrence, creating minor barriers for a competent reader from an adjacent field (e.g., a computer vision researcher not specializing in 3DGS or a geospatial analyst not familiar with specific rendering metrics).

Specifically, the acronym “DOF” in Table 1 is undefined. While “Degrees of Freedom” is a standard concept, the specific context of camera poses or scene parameters in this table requires explicit definition for clarity. Similarly, “Z-Monotonic SDF” in Section 2.1 is a specialized geometric constraint that is not standard vocabulary outside of specific reconstruction subfields; a brief explanation of what “Z-Monotonic” implies in this context would aid comprehension.

In the data quality and deployment sections, terms like “accumulated opacity” (Section 2.3), “Bhattacharyya distance” (Section 4.2), and “frustum culling” (Section 4.2) are used as if they are common knowledge. While these are standard in their respective niches (3DGS rendering, statistical distance metrics, and real-time graphics), they are not universally known across all adjacent fields. Defining these terms with a short clause or parenthetical expansion would significantly improve accessibility without diluting the technical precision.

Finally, the acronyms “ENU” (Section 4.2) and “KID” (Section 5.1) are introduced without expansion. “ENU” is a standard coordinate system in geodesy but should be spelled out for a general technical audience, and “KID” (Kernel Inception Distance) should be defined alongside FID to ensure the evaluation metrics are fully transparent.

Addressing these specific omissions will ensure the paper is self-contained for the target “adjacent-field PhD” audience, allowing them to follow the methodology and evaluation without needing to consult external glossaries or prior knowledge of specific 3DGS implementation details.

Action items.

- **[writing]** Section 2.1 (Data Collection): The acronym ‘DOF’ appears in Table 1 without definition. While ‘Degrees of Freedom’ is standard in robotics, it is not explicitly defined here. Add ‘(DOF: Degrees of Freedom)’ at first use in the table header or text.
- **[writing]** Section 2.1 (Data Collection): The term ‘Z-Monotonic SDF’ is introduced in the ‘Multi-Stereo Satellite Imagery’ paragraph without definition. A competent adjacent-field reader may not know this specific geometric constraint. Add a brief gloss, e.g., ‘a Signed Distance Field (SDF) constrained to be monotonic along the viewing ray’.
- **[writing]** Section 2.3 (Training Tile Generation): The term ‘accumulated opacity’ is used in ‘View-Level Rendering Assessment’ without definition. While intuitive to 3DGS experts, it is a specific technical term. Define it briefly, e.g., ‘the sum of opacities of all Gaussians along a ray’.

- **[writing]** Section 4.2 (EarthScape): The ‘Bhattacharyya distance’ is cited as the guide for statistical decimation without explanation. For a reader outside statistical learning or specific 3D compression subfields, this is opaque. Add a clause defining it as ‘a metric measuring the similarity of two probability distributions’.
- **[writing]** Section 4.2 (EarthScape): The acronym ‘ENU’ is used (‘ENU local tangent plane coordinate system’) without expansion. Define it as ‘(East-North-Up)’ at first occurrence.
- **[writing]** Section 4.2 (EarthScape): The term ‘frustum culling’ is used in the final paragraph without definition. While common in graphics, it is a specific rendering optimization. Add a brief parenthetical, e.g., ‘frustum culling (discarding objects outside the camera view)’.
- **[writing]** Section 5.1 (Generative Fidelity): The term ‘KID’ is used alongside FID without expansion. Define it as ‘Kernel Inception Distance (KID)’ at first use in the text.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a coherent argument for a generative 3D Earth model, but contains minor logical inconsistencies regarding numerical claims and scope definitions between sections.

1. Inference Rate Consistency Section 4.1 states a single inference pass takes ~25 minutes for a $1.6\text{km} \times 1.6\text{km}$ (2.56km^2) tile. Section 5.2 claims the system generates “1 km^2 in under 10 minutes.” While $25/2.56 = 9.8 \text{ min}/\text{km}^2$ mathematically supports the claim, the text presents the “1 km^2 ” metric as a standalone capability without explicitly linking it to the 2.56km^2 tile premise. This creates a slight gap where the reader must perform the division to verify the claim. The text should explicitly state “equivalent to under 10 minutes per km^2 ” to ensure the conclusion strictly follows the stated premise.

2. Precision of Speed Claims Table 1 (intro_compare.tex) lists the speed of “Ours” as “10 min,” whereas the Abstract and Section 5.2 consistently use “under 10 minutes.” Logically, “10 min” implies an exact value or upper bound of exactly 10, which contradicts the “under” qualifier if the actual time is 9.8. To maintain logical consistency, the table should reflect the bound (“<10 min”) or the text should be adjusted to “approximately 10 minutes.”

3. Scope of “Seamlessness” Section 3.3 claims the method ensures a “continuous user experience” and “vast, seamless landscapes.” However, Section 4.1 explicitly states that the current strategy focuses on seamlessness *within* each $1.6\text{km} \times 1.6\text{km}$ block and that “A fully seamless, cross-block inference strategy... is a key objective for future iterations.” The conclusion in Section 3.3 overreaches the current evidence by implying global seamlessness across all 312,500 production tiles, which the deployment section admits is not yet fully realized. The claim should be qualified to reflect that seamlessness is achieved within blocks, while cross-block continuity is handled by the tiling strategy with potential future improvements.

These issues are minor and fixable through precise rewording, but they currently represent small breaks in the logical chain between premises and conclusions.

Action items.

- **[writing]** Section 4.1 states a 25-min inference for a 2.56km² tile, while Section 5.2 claims <10 min/km². Clarify if the 10-min metric is a derived rate or a specific tile size to ensure the conclusion follows the premise.
- **[writing]** Table 1 lists speed as “10 min” while the text says “under 10 minutes”. Update the table to “<10 min” or the text to “approx. 10 min” to resolve the logical mismatch between the bound and the exact value.
- **[writing]** Section 3.3 claims “seamless” landscapes, but Section 4.1 admits cross-block seamlessness is a future goal. Qualify Section 3.3 to specify seamlessness is within blocks, not necessarily across the 312k production tiles.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several ambitious claims regarding the scope and capability of ABot-Earth 0.5 that exceed the evidence presented in the evaluation section. Specifically, the rhetoric frequently shifts from “demonstrating potential” to “solving” or “achieving” global capabilities without the corresponding empirical validation.

First, the Abstract and Introduction assert that the framework “effectively mitigates the sim-to-real domain gap” and enables “closed-loop UAV navigation.” However, Section 5 (Evaluation) only presents quantitative metrics (FID/KID) on 2D renderings and qualitative visual comparisons. There is no evidence of actual UAV navigation trials, control loop testing, or deployment in a physical or simulated robotic environment. The claim of enabling navigation is a projection of utility rather than a demonstrated result. The language should be hedged to reflect that the system *provides a simulation environment* for such tasks, rather than claiming to have *enabled* the tasks themselves.

Second, the Introduction claims “robust Earth-scale generalizability” across the “vast majority of global metropolitan areas as well as numerous non-urban natural terrains.” The evaluation in Section 5.1.2, however, relies on a very small set of qualitative examples (Auckland, Kamakura, Ireland) and a human study radar chart. This sample size is insufficient to support a claim of robust generalization across the “vast majority” of the globe. The claim should be narrowed to reflect the specific diversity of the tested cases or expanded with a more rigorous, geographically diverse quantitative benchmark.

Third, the paper claims to be the “first” to create “continuous, interactive, Earth-scale 3D environments.” Yet, Section 4.1 explicitly admits that the current production pipeline is

block-based (1.6km x 1.6km tiles) and that “A fully seamless, cross-block inference strategy is theoretically feasible and is a key objective for future iterations.” This admission directly contradicts the claim of “continuous” generation in the present tense. The claim of being the “first” should be qualified to acknowledge the current block-based nature of the implementation, or the “continuous” descriptor should be removed until cross-block seamlessness is empirically demonstrated.

Finally, Table 2 in the evaluation section lists the coverage as “Infinite.” Given the explicit description of a finite tile-based production pipeline with a specific count of ~312,500 tiles for the built-up area, “Infinite” is an overreach. “Global-scale” or “Planetary-scale” would be more accurate and scientifically precise terms that do not imply a mathematical infinity that the system’s finite compute resources cannot support.

These issues are primarily rhetorical overextensions of the current results. They can be resolved by tightening the language in the Abstract, Introduction, and Evaluation sections to align with the actual scope of the experiments and the admitted limitations of the current system architecture.

Action items.

- **[writing]** Abstract/Intro claim ‘solves’ sim-to-real gap and enables ‘closed-loop UAV navigation,’ but Sec 5 only shows static renderings/FID. No UAV trials exist. Replace ‘solves’ with ‘mitigates’ and scope navigation to ‘simulation-ready environments for’ until validated.
- **[writing]** Intro claims ‘robust Earth-scale generalizability’ across ‘vast majority’ of global areas, but Sec 5 only tests 3 locations (Auckland, Kamakura, Ireland). Narrow claim to ‘demonstrated on diverse test cases’ or provide broader quantitative evaluation across varied biomes.
- **[writing]** Table 1 and Sec 5 claim ‘first’ to create ‘continuous’ Earth-scale environments, yet Sec 4.1 admits cross-block seamlessness is a ‘future iteration’ goal. Qualify ‘first’ to ‘first pipeline for’ or acknowledge current block-based stitching limitations in the claim.
- **[writing]** Table 2 lists coverage as ‘Infinite,’ but Sec 4.1 describes a finite tile-based pipeline (~312k tiles) limited by VRAM. Replace ‘Infinite’ with ‘Global-scale’ or ‘Planetary-scale’ to accurately reflect finite compute constraints.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The paper presents a generative 3D Earth model trained on real-world reconstructions and satellite imagery. From a safety and ethics perspective, the work does not present a foreseeable, non-trivial risk of harm that is unaddressed.

The primary data sources are explicitly identified as public benchmarks (e.g., DFC 2019,

UrbanScene3D) and proprietary aerial/satellite imagery. The paper states that dynamic elements like vehicles and pedestrians are “automatically detected and removed” during the reconstruction pipeline (Section 2.2), which effectively mitigates risks related to PII or re-identification of individuals in the training data. The generated output is a synthetic 3D environment; while it is geospatially accurate, it does not expose sensitive personal data or operational details of critical infrastructure in a way that would facilitate immediate harm (e.g., it does not reveal internal layouts, security vulnerabilities, or real-time surveillance feeds).

The paper mentions downstream applications in UAV navigation and simulation. While generative models can theoretically be used for dual purposes, the method described (generating visual 3D scenes from satellite imagery) does not lower the barrier to a specific harmful capability (such as automated vulnerability discovery or biological synthesis) in a way that requires a novel mitigation strategy beyond standard responsible AI practices. The authors acknowledge the “sim-to-real” gap and the utility for simulation, which is a standard and acceptable research goal.

No human-subjects research requiring IRB approval is described (the data is pre-existing imagery or synthetic reconstructions). No license violations are apparent given the mix of public datasets and proprietary data. The paper does not disclose any conflicts of interest that would bias the safety assessment, though the authors are affiliated with a commercial mapping entity (Amap/AMAP), which is standard for this type of applied research and does not constitute an undisclosed conflict in this context.

Consequently, there are no specific, nameable gaps in disclosure or mitigation that require action. The work is low-risk by construction.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The central claim of ABot-Earth 0.5 is that it achieves state-of-the-art generative fidelity and system-level applicability for planetary-scale 3D reconstruction. However, the experimental design presented in the Evaluation section contains critical gaps that prevent the evidence from supporting these claims.

First, the quantitative comparison of generative fidelity (Table 1) is fundamentally flawed. The authors report an FID of 16.1 compared to baselines ranging from 69.5 to 97.3. However, the table caption explicitly states: “FID/KID values for baselines are computed using different GT sets than ours. In addition, the poses/viewpoints used for evaluation differ across methods.” This admission invalidates the comparison. The observed gap could be entirely explained by the baselines being evaluated on easier or different data distributions rather than the proposed method being superior. Without a unified evaluation protocol where all methods are rendered from identical camera poses against the same ground-truth distribution, the claim of “state-of-

the-art” performance is unsupported.

Second, the human study comparing visual quality against Google Earth (Section 5.2.3) lacks a necessary negative control. The study asks users to rate “Geometric Accuracy,” “Textural Fidelity,” and “Aesthetics.” However, Google Earth represents a photogrammetric reconstruction of the real world (ground truth), while ABot-Earth is a generative model. The design fails to distinguish whether users prefer the “plausible hallucinations” of the generative model or if the generative model simply benefits from smoother, less noisy textures that are easier to rate highly in a blind test. To isolate the contribution of the generative method, the study should include a baseline that uses a downsampled or simplified version of the actual Google Earth reconstruction (or a similar real-world scan) to determine if the generative model truly adds value or merely masks reconstruction artifacts.

Third, the efficiency claims are conflated. The paper highlights a generation rate of “under 10 minutes per square kilometer” as a key advantage. However, Section 4.1 describes a massive production run using 1,000 GPUs over 10 days to cover 800,000 km². The design does not isolate the latency of a single inference pass on a standard hardware configuration. The 10-minute figure likely relies on the massive parallelization of the 1,000-GPU cluster, which does not reflect the cost or time for a single user or smaller deployment. The claim of “ultra-low-cost” and “high-efficiency” is not supported without reporting the single-tile inference time on a standard single-GPU setup.

Finally, the paper attributes the high quality of the output to the “native 3DGS generative framework.” However, the training data is entirely derived from the authors’ own ABot-3DGS reconstruction engine. There is no ablation study that isolates the generative model’s contribution from the quality of the training data pipeline. It is plausible that the high fidelity comes from the superior reconstruction capabilities of ABot-3DGS rather than the generative model itself. A control experiment replacing the generative model with a simple retrieval or interpolation of training tiles would be necessary to prove the generative component is the source of the improvement.

These design flaws mean the current evidence cannot distinguish the proposed method’s specific contributions from confounding factors like evaluation bias, data pipeline quality, or hardware scaling.

Action items.

- **[science]** The central claim of ABot-Earth 0.5 is that it achieves state-of-the-art generative fidelity and system-level applicability for planetary-scale 3D reconstruction. However, the experimental design presented in the Evaluation section contains critical gaps that prevent the evidence from supporting these claims. First, the quantitative comparison of generative fidelity (Table 1) is fundamentally flawed. The authors report an FID of 16.1 compared to baselines ranging from 69.5 to 97.3. However, the

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical treatment of the quantitative results in this paper is largely descriptive but lacks the inferential rigor required to support claims of “significance” or robust superiority.

First, **Table 1** (tab:conditions_results) presents FID and KID scores as single point estimates (e.g., 16.1 for Ours). While the caption notes that baseline metrics are cited from other works (potentially single runs), the proposed method’s result is presented without any measure of variance (standard deviation, standard error, or confidence intervals). In generative modeling, FID scores can fluctuate significantly based on random seeds, batch sampling, or initialization. Reporting a single number implies a level of precision that is not statistically justified. The authors should report the mean and standard deviation of the FID/KID scores over at least 3-5 independent training runs (seeds) to demonstrate the stability of the result.

Second, the **human study** described in Section 5.1.3 (“Visual Quality and Aesthetics”) is statistically opaque. The text claims the method achieves a “higher aesthetic score” based on a radar chart (Fig. 5), but provides no data on the number of participants, the statistical test used to compare scores, or measures of inter-rater reliability. Without a sample size (N) and a p-value or confidence interval, the assertion that the aesthetic score is higher is anecdotal rather than empirical. The authors must report the number of raters, the statistical test employed (e.g., paired t-test or Wilcoxon signed-rank test), and the resulting significance levels to validate this claim.

Finally, the use of the term “**significantly better**” in the Abstract and Introduction is unsupported by the evidence presented in Section 5.1.1. The section compares point estimates (16.1 vs 69.5) but does not perform or report a hypothesis test. In statistical reporting, “significant” implies a specific threshold (e.g., $p < 0.05$) derived from a test. If no test was run, the language should be softened to “substantially lower” or “improved,” or the authors must conduct the appropriate statistical test (e.g., bootstrapping the FID distribution) and report the p-value.

These are primarily reporting and analysis gaps (writing/science) rather than fatal flaws in the experimental design, but they currently prevent the quantitative claims from being fully trustworthy.

Action items.

- **[writing]** Table 1 (tab:conditions_results) reports FID/KID scores for baselines (CityDreamer, GaussianCity, EarthCrafter) as single point estimates without uncertainty metrics (e.g., mean $\pm$ SD over seeds). The caption admits baselines use different GT sets, but the lack of variance reporting for the proposed method’s 16.1 FID score prevents assessment of stability. Report mean $\pm$ SD over 3 random seeds for the proposed method and clarify if baseline numbers are single-run citations or aggregated.
- **[science]** Section 5.1.3 claims a ‘comprehensive human study’ for visual quality (radar

chart in Fig. 5) but provides no statistical details: sample size (N participants), inter-rater reliability (e.g., Cronbach’s alpha), or significance tests (e.g., paired t-tests/ANOVA) for the claimed ‘higher aesthetic score.’ Without these, the claim of superiority is anecdotal. Add N, reliability metrics, and p-values or confidence intervals for the human study results.

- **[writing]** The abstract and Section 1 claim the method is ‘significantly better’ than baselines, yet Section 5.1.1 only presents point estimates (FID 16.1 vs 69.5) without a formal hypothesis test or confidence intervals to support the ‘significant’ descriptor. In the absence of a stated test (e.g., bootstrap or t-test) and p-value, replace ‘significantly better’ with ‘substantially lower FID’ or report the actual statistical test results.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The paper presents a clear and compelling narrative, effectively guiding the reader from the limitations of current 3D reconstruction methods to the proposed generative solution. The abstract successfully summarizes the method, results, and implications. However, there are specific areas where sentence construction and terminology consistency create minor friction.

In Section 4, the description of the georeferencing preprocessing contains slightly ambiguous phrasing. The phrase “isotropic resampling process based on the target spatial extent” requires the reader to infer the reference point. Explicitly stating that this refers to the “original ground coverage” would resolve this immediately.

In Section 5, there is a minor inconsistency between the evaluation dimensions listed in the introduction and the actual subsection titles. The introduction lists “Timeliness and Efficiency” and “Visual Quality,” while the subsections are titled “Efficiency” and “Visual Quality and Aesthetics.” Aligning these titles would improve the structural coherence of the section.

Additionally, in Section 3, the use of curly braces {} for emphasis in subsection titles (e.g., {representation gap}) is non-standard LaTeX practice and may cause rendering issues. Standard italics or bolding should be used instead. Finally, the introduction of the “FromOrbit2Ground” module in Section 2 is slightly dense; breaking the definition and its function into two distinct sentences would enhance readability.

These issues are minor and do not obscure the scientific contribution, but addressing them will polish the prose to match the high quality of the research.

Action items.

- **[writing]** Section 4, ‘Georeferencing’: The phrase ‘isotropic resampling process based on the target spatial extent’ is vague. Clarify that ‘target spatial extent’ refers to the original ground coverage to ensure the resampling logic is immediately clear.
- **[writing]** Section 5: The subsection titles (‘Efficiency’, ‘Visual Quality and Aesthetics’) do

not match the introductory list ('Timeliness and Efficiency', 'Visual Quality'). Align the titles with the list to improve structural coherence.

- **[writing]** Section 3: Subsection titles use curly braces for emphasis (e.g., '{representation gap}'), which is non-standard. Replace with italics or bold text to ensure clean rendering and professional appearance.
- **[writing]** Section 2, 'Multi-Stereo Satellite Imagery': The definition of 'FromOrbit2Ground' is dense. Split the sentence introducing it and its function into two sentences to improve flow and clarity.